SAT Math Final Revision Set 1

76 questions · 2 hours

Name ..... Date .....

Each question states its tool: **No calculator needed**, **Calculator** or **Desmos**. Multiple-choice questions have one correct answer. For student-produced responses (SPR), write your answer in the box.

Algebra (13 items)

1. A-316 · Hard · SPR · **No calculator needed**

The function f is linear, $f(3) = k$ and $f(k) = 3$, where k is a constant and $k \neq 3$. If $f(10) = -1$, what is the value of k?

2. A-318 · Hard · SPR · **No calculator needed**

The function f is linear, $f(2) + f(8) = 30$, and $f(11) = 33$. What is the value of $f(0)$?

3. A-321 · Hard · Multiple choice · **No calculator needed**

The system $2y - x > 4$ and $y < 5$ is graphed in the xy-plane. Which inequality describes all possible x-coordinates of points in the solution region?

A) $x > 6$

B) $x < 14$

C) $x < 6$

D) $x < -4$

4. A-323 · Hard · SPR · **No calculator needed**

If $-3 \leq x \leq 2$ and $-5 \leq y \leq 1$, what is the greatest possible value of xy?

5. A-328 · Hard · SPR · **No calculator needed**

In the system $y = 2x + k$ and $y = -x + 10$, k is a constant. For how many integer values of k is the solution (x, y) in Quadrant I (that is, $x > 0$ and $y > 0$)?

6. **A-332** · Hard · Multiple choice · **No calculator needed**

Consider the system $3x - 4y = 10$ and $-6x + 8y = -20$. Which point lies on the graphs of both equations for every real number t?

A) $\left(t, \frac{3t - 10}{4}\right)$

B) $\left(t, \frac{10 - 3t}{4}\right)$

C) $\left(t, \frac{3t + 10}{4}\right)$

D) $\left(\frac{4t - 10}{3}, t\right)$

7. **A-335** · Hard · SPR · **No calculator needed**

An oven's temperature is $70°$F when it is switched on at $t = 0$. For $3 \leq t \leq 12$, where t is the time in minutes, the temperature is modeled by $T = a + bt$, where a and b are constants. The model gives $T = 200$ at $t = 3$ and $T = 380$ at $t = 7$. What is the value of a?

8. **A-338** · Hard · Multiple choice · **No calculator needed**

In the xy-plane, the graph of $3x - ky = 12$, where k is a nonzero constant, has an x-intercept equal to its y-intercept. What is the value of k?

A) -4

B) 3

C) 4

D) -3

9. **A-340** · Hard · Multiple choice · **No calculator needed**

A cafe sells juice for 4 dollars each and sandwiches for 7 dollars each. A customer spends exactly 112 dollars and buys at least one juice and at least one sandwich. How many different combinations of juices and sandwiches are possible?

A) 5

B) 4

C) 3

D) 28

10. **A-319** · Medium · Multiple choice · **No calculator needed**

In the xy-plane, what is the y-intercept of the graph of $x = 4y - 12$?

A) $(0, -12)$

B) $(0, -3)$

C) $(-12, 0)$

D) $(0, 3)$

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11. **A-325** · Medium · Multiple choice · **No calculator needed**

A line in the xy-plane has slope 3 and passes through the points $(2, k)$ and $(k, 14)$. What is the value of k?

A) 8

B) 4

C) 11

D) 5

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12. **A-329** · Medium · Multiple choice · **No calculator needed**

In the inequality $kx + 4 < 10$, k is a constant and $k < 0$. Which describes the solution set?

A) $x < \frac{6}{k}$

B) $x > \frac{6}{k}$

C) $x > -\frac{6}{k}$

D) $x < -\frac{6}{k}$

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13. **A-334** · Medium · Multiple choice · **No calculator needed**

How many solutions does the system $2x + y = 11$, $x - y = 1$, $x + 3y = -13$ have?

A) Exactly one

B) Exactly three

C) Zero

D) Exactly two

Advanced Math (15 items)

14. **AM-309** · Hard · Multiple choice · **Desmos**

How many solutions (x, y) does the system $x^2 + y^2 = 25$ and $y = x^2 - 5$ have?

A) 3

B) 4

C) 1

D) 2

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15. **AM-317** · Hard · Multiple choice · **Desmos**

In the xy-plane, the graphs of $y = |x - 2|$ and $y = mx + 1$, where m is a constant, intersect at exactly two points. Which of the following could be the value of m?

A) -2

B) $-\frac{3}{4}$

C) 2

D) $\frac{1}{2}$

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16. **AM-303** · Hard · Multiple choice · **No calculator needed**

In the equation $(k - 2)x^2 + 4x + 1 = 0$, k is a constant. The equation has exactly one real solution for exactly two values of k. What is the sum of these two values of k?

A) 8

B) 4

C) 2

D) 6

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17. **AM-305** · Hard · Multiple choice · **No calculator needed**

The function h is defined by $h(x) = \frac{x^2 - 4x - 12}{x^2 - 36}$. What is the sum of all values of x for which $h(x) = 0$?

A) 4

B) -8

C) 6

D) -2

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18. **AM-307** · Hard · Multiple choice · **No calculator needed**

In the equation $\frac{x^2}{x-a} = \frac{9}{x-a}$, a is a constant. For how many real values of a does the equation have exactly one real solution?

A) Infinitely many

B) Exactly two

C) Exactly one

D) Zero

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19. **AM-312** · Hard · Multiple choice · **No calculator needed**

How many real solutions does the equation $\sqrt{x^2 - 10x + 25} = 5 - x$ have?

A) Exactly two

B) Zero

C) Infinitely many

D) Exactly one

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20. **AM-314** · Hard · Multiple choice · **No calculator needed**

The expression $\frac{6x^2 + 7x - 5}{2x + 3}$ is equivalent to $Ax + B + \frac{C}{2x + 3}$, where A, B and C are constants. What is the value of $A + B + C$?

A) -4

B) 2

C) 4

D) 0

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21. **AM-316** · Hard · Multiple choice · **No calculator needed**

For all real numbers x, $f(2x - 1) = 4x^2 + 3$. Which of the following defines $f(x)$?

A) $f(x) = x^2 + 4$

B) $f(x) = x^2 + 2x + 4$

C) $f(x) = 4x^2 + 3$

D) $f(x) = 4x^2 - 4x + 4$

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22. **AM-320** · Hard · SPR · **No calculator needed**

In the equation $\frac{x+a}{x^2-a^2} = -\frac{3}{20x}$, a is a constant. If $x = 6$ is a solution, what is the value of a?

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23. **AM-322** · Hard · Multiple choice · **No calculator needed**

In the equation $\frac{ax}{x-2} = 3 + \frac{4}{x-2}$, a is a constant. For which values of a does the equation have no solution?

A) 3 and 4 only

B) 3 only

C) 2 only

D) 2 and 3 only

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24. **AM-324** · Hard · SPR · **No calculator needed**

For how many integer values of b does the equation $x^2 + bx + 36 = 0$ have two distinct integer solutions?

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25. **AM-327** · Hard · Multiple choice · **No calculator needed**

What is the solution set of the equation $2x^{2/3} - 3 = 15$?

A) $\{-27, 27\}$

B) $\{27\}$

C) $\{-9^{2/3}, 9^{2/3}\}$

D) $\{-6, 6\}$

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26. **AM-329** · Hard · Multiple choice · **No calculator needed**

The function f is defined by $f(x) = 2(x-1)(x-7)$. For what value of k does the graph of $y = f(x) + k$ have exactly one x-intercept?

A) 9

B) 14

C) 18

D) -18

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27. **AM-311** · Medium · SPR · **No calculator needed**

What is the sum of all solutions to the equation $(x-3)^2 = (2x+1)^2$?

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28. **AM-331** · Medium · Multiple choice · **No calculator needed**

If $2a - 3b = 7$, what is the value of $\frac{4^a}{8^b}$?

A) $\frac{1}{128}$

B) 2^{14}

C) 7

D) 128

Geometry and Trigonometry (21 items)

29. **GT-255** · Hard · Multiple choice · **Calculator**

Model Earth as a sphere with radius $6{,}371$ kilometers. An observer's eye is 1 kilometer above the surface. The line of sight from the eye to the horizon is tangent to Earth. Which is closest to the distance, in kilometers, from the eye to the point where the line of sight touches Earth?

A) 112.9

B) 1.0

C) $6{,}371.0$

D) 79.8

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30. **GT-232** · Hard · Multiple choice · **No calculator needed**

In the xy-plane, a circle has equation $(x-4)^2 + (y+2)^2 = 50$. Which point lies in the interior of the circle?

A) $(10, 2)$

B) $(9, 3)$

C) $(-1, -6)$

D) $(-3, 0)$

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31. GT-234 · Hard · Multiple choice · **No calculator needed**

For $0 < t < \frac{\pi}{2}$, $\sin t = 0.6$. What is the value of $\cos(\pi + t) + \sin(\pi - t)$?

A) 0.2

B) -1.4

C) -0.2

D) 1.4

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32. GT-236 · Hard · Multiple choice · **No calculator needed**

In triangle ABC, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that the measure of angle AED equals the measure of angle ABC. If $AD = 4$, $DB = 11$ and $AE = 6$, what is the length of $\overline{EC}$?

A) 10

B) 16.5

C) 22.5

D) 4

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33. GT-239 · Hard · SPR · **No calculator needed**

A rhombus has side length 13 and area 120. What is the sum of the lengths of its diagonals?

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34. GT-241 · Hard · SPR · **No calculator needed**

A sector of a circle has radius 10 and central angle $216°$. The sector is rolled into a right circular cone so that its two straight edges meet. The volume of the cone is $k\pi$. What is the value of k?

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35. GT-243 · Hard · Multiple choice · **No calculator needed**

In triangles PQR and XYZ, $PQ = XY = 9$ and $m\angle R = m\angle Z = 35°$, where $\overline{PQ}$ is opposite angle R and $\overline{XY}$ is opposite angle Z. Which additional fact is enough to prove that triangle PQR is congruent to triangle XYZ?

A) $PR = XZ$

B) $QR = YZ = 14$

C) $m\angle Q = m\angle Y$

D) $PR = QR$

36. GT-246 · Hard · Multiple choice · **No calculator needed**

Triangle ABC has area 60. Point D lies on $\overline{BC}$ with $BD : DC = 2 : 3$, and point E lies on $\overline{AD}$ with $AE : ED = 3 : 1$. What is the area of triangle BED?

A) 15

B) 18

C) 10

D) 6

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37. GT-249 · Hard · Multiple choice · **No calculator needed**

Sphere Q has volume 36π. Sphere P has a radius 7 times the radius of sphere Q. The spheres are externally tangent at point T. Segment $\overline{MT}$ is a diameter of sphere P and segment $\overline{TN}$ is a diameter of sphere Q, with M, T and N on one line. What is the length of $\overline{MN}$?

A) 66

B) 45

C) 24

D) 48

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38. GT-252 · Hard · SPR · **No calculator needed**

Two identical right rectangular prisms each have a square base with side length s and a height of 35. Each prism has surface area K. The prisms are glued together base to base, and the new solid has surface area $\frac{15}{8}K$. What is the value of s?

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39. GT-256 · Hard · Multiple choice · **No calculator needed**

Parallel lines m and n are cut by a transversal. Angles b and c are same-side interior angles, and angle a is one of the other angles formed. If $200 < a + b + c < 236$, where all measures are in degrees, which of the following could be the measure of angle a, in degrees?

A) 56

B) 18

C) 64

D) 40

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40. GT-258 · Hard · Multiple choice · **No calculator needed**

Right triangles PQR and XYZ are similar, with P, Q and R corresponding to X, Y and Z, and right angles at Q and Y. If $\tan P = \dfrac{8}{15}$ and $XZ = 51$, what is the length of $\overline{XY}$?

A) 27.2

B) 45

C) 24

D) 95.625

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41. GT-260 · Hard · Multiple choice · **No calculator needed**

A rectangular box has a base that measures 9 by 12 and a height of 20. Let θ be the angle between a diagonal of the box (from a bottom corner to the opposite top corner) and the base. What is the value of $\tan\theta$?

A) $\dfrac{20}{9}$

B) $\dfrac{5}{3}$

C) $\dfrac{3}{4}$

D) $\dfrac{4}{3}$

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42. GT-262 · Hard · Multiple choice · **No calculator needed**

Right triangle ABC has its right angle at C, $AC = 9$ and $BC = 12$. Each leg is lengthened by 3 and the angle at C stays a right angle. What is $\tan A$ in the new triangle?

A) $\dfrac{4}{5}$

B) $\dfrac{4}{3}$

C) $\dfrac{5}{4}$

D) $\dfrac{5}{3}$

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43. GT-264 · Hard · Multiple choice · **No calculator needed**

A drone flies 12 kilometers in a direction 60° north of east, then flies 12 kilometers due west. How far, in kilometers, is the drone from its starting point?

A) 12

B) 24

C) $12\sqrt{2}$

D) $6\sqrt{3}$

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44. GT-266 · Hard · Multiple choice · **No calculator needed**

In the xy-plane, the points $A(0,0)$, $B(6,3)$ and $C(4,7)$ are the vertices of a triangle. What is the value of $\tan(\angle CAB)$?

A) $\frac{1}{2}$

B) $\frac{2}{3}$

C) $\frac{3}{2}$

D) $\frac{7}{4}$

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45. GT-269 · Hard · Multiple choice · **No calculator needed**

In a right triangle, the measure of acute angle A is between $45°$ and $90°$. Which of the following is true?

A) $\cos A < \sin A < \tan A$

B) $\tan A < \sin A < \cos A$

C) $\sin A < \cos A < \tan A$

D) $\cos A < \tan A < \sin A$

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46. GT-271 · Hard · Multiple choice · **No calculator needed**

In the xy-plane, line ℓ has slope $\frac{3}{4}$ and line k is perpendicular to line ℓ. Line k makes an angle θ with the positive x-axis, measured counterclockwise, where $0° < \theta < 180°$. What is the value of $\cos\theta$?

A) $-\frac{4}{5}$

B) $\frac{3}{5}$

C) $-\frac{3}{5}$

D) $\frac{4}{5}$

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47. GT-273 · Hard · Multiple choice · **No calculator needed**

In triangle ABC, the measure of angle A is $52°$. The bisectors of angles B and C intersect at point I. What is the measure, in degrees, of angle BIC?

A) 116

B) 128

C) 64

D) 104

48. **GT-238** · Medium · Multiple choice · **No calculator needed**

A triangle has two sides of lengths 10 and 12, and the angle between them measures $150°$. What is the area of the triangle?

A) $30\sqrt{3}$

B) 30

C) 15

D) 60

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49. **GT-267** · Medium · Multiple choice · **No calculator needed**

A parallelogram has sides of lengths 12 and 20. The cosine of one of its acute angles is 0.8. What is the area of the parallelogram?

A) 72

B) 144

C) 192

D) 240

Problem-Solving and Data Analysis (27 items)

50. **PS-164** · Hard · Multiple choice · **Calculator**

Electric vehicles made up 4% of a country's new-car sales in one year and 6% the next year. Total new-car sales increased by 10% from the first year to the second. By what percent did the number of electric vehicles sold increase?

A) 2%

B) 60%

C) 65%

D) 50%

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51. **PS-167** · Hard · SPR · **Calculator**

Bottle B holds 20% more juice than bottle A and costs 15% more. To the nearest tenth of a percent, by what percent is the price per liter of bottle B less than the price per liter of bottle A?

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52. PS-179 · Hard · Multiple choice · **Calculator**

A restaurant adds a 15% tax and a 10% service charge to a bill. Both are calculated on the original subtotal. The total paid is 250 dollars. What was the subtotal, in dollars?

A) 200.00

B) 217.39

C) 197.63

D) 187.50

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53. PS-182 · Hard · Multiple choice · **Calculator**

Two positive numbers a and b are in the ratio $a : b = 9 : 5$. The number b is what percent less than a, to the nearest tenth of a percent?

A) 44.4%

B) 80.0%

C) 55.6%

D) 28.6%

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54. PS-157 · Hard · Multiple choice · **No calculator needed**

Data sets A and B each contain 20 values, all from the list $10, 20, 30, 40, 50$. The frequencies in A are $2, 3, 10, 3, 2$ and the frequencies in B are $5, 3, 4, 3, 5$, in that order. Which statement is true?

A) The standard deviations are equal because the means and ranges are equal.

B) The standard deviation of B is greater.

C) The standard deviation of A is greater.

D) The standard deviations cannot be compared without calculating them.

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55. PS-160 · Hard · SPR · **No calculator needed**

The data set $2, 6, 9, 15, x$ has a mean equal to its median, where x is a real number. What is the sum of all possible values of x?

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56. PS-162 · Hard · Multiple choice · **No calculator needed**

The commute times, in whole minutes, of 20 students are grouped: 10 students took 0 to 9 minutes, 7 took 10 to 19 minutes, and 3 took 20 to 29 minutes. Which of the following could NOT be the median commute time?

A) 10

B) 7.5

C) 14

D) 4

57. **PS-165** · Hard · SPR · **No calculator needed**

The number a is 150% more than the number b, and b is 60% less than the positive number c. The number a is what percent of c?

58. **PS-169** · Hard · SPR · **No calculator needed**

In grade 10, 12 students are in a club and 28 are not. In grade 11, k students are in the club and 42 are not. Being in the club is independent of grade. What is the value of k?

59. **PS-171** · Hard · SPR · **No calculator needed**

A line of best fit for a data set is $\hat{y} = 2.5x + 40$. For the data point with $x = 24$, the predicted value of y is 20% less than the actual value. What is the residual for this data point?

60. **PS-174** · Hard · SPR · **No calculator needed**

A line of best fit for a data set is $\hat{y} = 2.4x + 15$. A new data set is formed by adding 5 to every x-value and keeping each y-value. What is the y-intercept of the line of best fit for the new data set?

61. **PS-176** · Hard · Multiple choice · **No calculator needed**

If 18% of a is 24, what is $a\%$ of 18?

A) 4.32

B) 24

C) $2{,}400$

D) $133\frac{1}{3}$

62. **PS-183** · Hard · Multiple choice · **No calculator needed**

The number y is $p\%$ less than the positive number x, where $0 < p < 100$. Which expression is equal to x?

A) $\dfrac{(100-p)y}{100}$

B) $\dfrac{100y}{100+p}$

C) $y\left(1+\dfrac{p}{100}\right)$

D) $\dfrac{100y}{100-p}$

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63. **PS-185** · Hard · Multiple choice · **No calculator needed**

In a survey, exactly 37.5% of the people answered "yes," exactly 20% answered "no," and the rest were undecided. What is the least possible number of people surveyed?

A) 5

B) 40

C) 8

D) $1{,}000$

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64. **PS-188** · Hard · Multiple choice · **No calculator needed**

A pipe can fill an empty tank in 6 hours, and a drain can empty the full tank in 9 hours. Starting with the tank empty, the pipe and the drain are both open for 4 hours, and then the drain is closed. How many more hours does it take the pipe to fill the tank?

A) 2

B) $\dfrac{26}{3}$

C) $\dfrac{14}{3}$

D) 14

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65. **PS-190** · Hard · Multiple choice · **No calculator needed**

A driver travels from one city to another at an average speed of 60 kilometers per hour. At what average speed, in kilometers per hour, must the driver return along the same road so that the average speed for the whole round trip is 80 kilometers per hour?

A) 120

B) 106.7

C) 160

D) 100

66. **PS-193** · Hard · Multiple choice · **No calculator needed**

The data set $3, 5, 6, 8, 9, 11, 12, 46$ has one outlier, 46, which is removed. Which statement about the effect of removing 46 is true?

A) Only the mean changes.

B) Both the mean and the median decrease, and the mean decreases more.

C) The mean decreases and the median increases.

D) The mean decreases and the median stays the same.

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67. **PS-195** · Hard · Multiple choice · **No calculator needed**

Section P has 60 students and section Q has 24 students. Test-score summaries: Section P: minimum 55, first quartile 68, median 76, third quartile 84, maximum 97. Section Q: minimum 62, first quartile 77, median 83, third quartile 88, maximum 95. Which statement must be true?

A) At least 75% of the students in Q scored above the median of P.

B) The IQR of Q is greater than the IQR of P.

C) More students in Q than in P scored above 84.

D) The range of Q is greater than the range of P.

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68. **PS-197** · Hard · SPR · **No calculator needed**

In a data set, the values 1, 2, 3, 4 and 5 occur with relative frequencies 10%, 15%, 20%, 30% and 25%, respectively. What is the median of the data set minus the mean of the data set?

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69. **PS-199** · Hard · Multiple choice · **No calculator needed**

A data set of 30 values has a mean of 50 and a standard deviation of 10. The values 38 and 62 are added to the data set. Which statement is true?

A) The mean is unchanged and the standard deviation increases.

B) The mean is unchanged and the standard deviation decreases.

C) The mean and the standard deviation are both unchanged.

D) The mean and the standard deviation both increase.

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70. **PS-202** · Hard · Multiple choice · **No calculator needed**

A company randomly selected 200 of its employees and gave them new office chairs. Their average productivity rose 8% during the next 3 months. Which conclusion is best supported?

A) The new chairs caused the rise for all employees.

B) The rise cannot be attributed to the chairs, because there was no comparison group.

C) The new chairs had no effect on productivity.

D) The new chairs caused the rise for the 200 employees only.

71. **PS-204** · Hard · Multiple choice · **No calculator needed**

A survey was mailed to $1{,}200$ randomly selected households, and 180 surveys were returned. Of these, 72% supported a new fee. What is the main reason the 72% may not represent all households?

A) There was no random assignment to treatments.

B) The households were selected at random.

C) The sample of $1{,}200$ is too small.

D) Many selected households did not respond, and those who responded may differ from those who did not.

72. **PS-210** · Hard · Multiple choice · **No calculator needed**

A model for a data set is $\hat{y} = 2.5x + 10$. The data points are $(2, 17)$, $(4, 19.5)$, $(6, 24)$, $(8, 31)$ and $(10, 34.5)$. For how many of the data points does the model underestimate the actual value of y?

A) 5

B) 2

C) 3

D) 1

73. **PS-187** · Medium · Multiple choice · **Calculator**

A runner's pace is 6 minutes 15 seconds per kilometer. What is the runner's speed, in kilometers per hour?

A) 9.6

B) 6.25

C) 9.76

D) 0.16

74. **PS-173** · Medium · SPR · **No calculator needed**

The ratio of grade 9 students to grade 10 students in a club is $3 : 5$, and the ratio of grade 10 students to grade 11 students is $4 : 7$. The club has 201 students in grades 9, 10 and 11. How many more grade 11 students than grade 9 students are in the club?

75. **PS-205** · Medium · Multiple choice · **No calculator needed**

An observational study found that children from homes with more books tend to have higher reading scores. Which of the following is the most plausible confounding variable?

A) Parents' education level

B) The scale used for the reading test

C) The number of pages per book

D) The number of children in the study

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76. **PS-207** · Medium · Multiple choice · **No calculator needed**

A teacher randomly selected 50 students and asked each one, face to face, "Have you ever copied homework?" Of the students, 6% answered yes. Which is the most likely problem with this result?

A) The sample was not selected at random.

B) There was no control group.

C) Students may give untrue answers because the question is sensitive and asked in person.

D) A sample of 50 students is too large.
